

8 Queens Problem

Python Code:

N = 8

```
def print_solution(board):
    for row in board:
        print(" ".join("Q" if col else "." for col in row))
    print("\n")

def is_safe(board, row, col):
    for i in range(row):
        if (
            board[i][col] or
            (col - (row - i) >= 0 and board[i][col - (row - i)]) or
            (col + (row - i) < N and board[i][col + (row - i)])
        ):
            return False
    return True

def solve_n_queens(board, row):
    if row == N:
        print_solution(board)
        return True

    found = False
    for col in range(N):
        if is_safe(board, row, col):
            board[row][col] = 1
            found = solve_n_queens(board, row + 1) or found
            board[row][col] = 0

    return found

board = [[0] * N for _ in range(N)]
solve_n_queens(board, 0)
```

OUTPUT:

```
main.py          [Icons] [Share] [Run] [Output] [Clear]
4
3 def print_solution(board):
4     for row in board:
5         print(" ".join("Q" if col else "." for col in row))
6     print("\n")
7
8 def is_safe(board, row, col):
9     for i in range(row):
10        if (
11            board[i][col] or
12            (col - (row - i) >= 0 and board[i][col - (row - i)]) or
13            (col + (row - i) < N and board[i][col + (row - i)])
14        ):
15            return False
16    return True
17
18 def solve_n_queens(board, row):
19     if row == N:
20         print_solution(board)
21         return True
22
23     found = False
24     for col in range(N):
25         if is_safe(board, row, col):
26             board[row][col] = 1
27             found = solve_n_queens(board, row + 1) or found
28             board[row][col] = 0
29
30     return found
31
32 board = [[0] * N for _ in range(N)]
33 solve_n_queens(board, 0)
```